RAJALAKSHMI ENGINEERING COLLEGE

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J2

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|---|---|
| **Started on** | Saturday, 3 October 2026, 4:26 PM |
| **State** | Finished |
| **Completed on** | Saturday, 3 October 2026, 4:30 PM |
| **Time taken** | 3 mins 49 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

Given two strings find the length of the common longest subsequence(need not be contiguous) between the two.

Example:

s1: ggtabe

s2: tgatasb

| | | | | | | |
|---|---|---|---|---|---|---|
| s1 | a | g | **g** | **t** | **a** | **b** |
| s2 | **g** | x | **t** | **x** | **a** | **y** | **b** |

**The length is 4**

Solveing it using Dynamic Programming

**For example:**

| Input | Result |
|---|---|
| aab<br>azb | 2 |

**Answer:** (penalty regime: 0 %)

```c
#include <stdio.h>
#include <string.h>

int main() {
    char s1[100], s2[100];

    scanf("%s", s1);
    scanf("%s", s2);

    int m = strlen(s1);
    int n = strlen(s2);

    int dp[m + 1][n + 1];

    // Initialize first row and first column
    for (int i = 0; i <= m; i++) {
        for (int j = 0; j <= n; j++) {
            if (i == 0 || j == 0) {
                dp[i][j] = 0;
            }
        }
    }

    // Fill DP table
    for (int i = 1; i <= m; i++) {
        for (int j = 1; j <= n; j++) {

            if (s1[i - 1] == s2[j - 1]) {
                dp[i][j] = dp[i - 1][j - 1] + 1;
            }
            else {
                if (dp[i - 1][j] > dp[i][j - 1])
                    dp[i][j] = dp[i - 1][j];
                else
                    dp[i][j] = dp[i][j - 1];
            }
        }
    }

    printf("%d", dp[m][n]);

    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | aab<br>azb | 2 | 2 | ✔ |
| ✔ | ABCD<br>ABCD | 4 | 4 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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